

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)			Box under 1 st diagram ticked (1)	1			1		
	(b)	(i)		Any 1 (1) from - Reverse field [accept: swap magnets; not just change field] - reverse current [accept: change current <u>direction</u>] - reverse voltage / switch around power supply	1			1		
		(ii)		Stronger {field / magnets} [accept: move poles closer together / add iron core] [not: more magnets] (1) Larger current / voltage (1) [power = neutral] Not: reference to area or turns on coil	2			2		
	(c)	(i)		50 [J]		1		1	1	
		(ii)		Substitution: $\left(\frac{50 \text{ ecf}}{200}\right) [\times 100]$ (1) ecf only allowed on $E_{\text{in}} < 200$ % efficiency = 25 (1) NB. 0.25 on answer line → substitution mark awarded	1	1		2	2	
				Question 4 total	5	2	0	7	3	0